

Ripasso derivate comuni in statistica

Esercizi

1.

$$f(x) = x^2 e^{-x} + 3 \ln(x^2 - 1) - \frac{\ln(x)}{x}$$

Sol.

$$f'(x) = 2x e^{-x} + x^2 (-e^{-x}) + 3 \frac{1}{x^2 - 1} 2x - \frac{\frac{1}{x} x - \ln(x)}{x^2}$$

$$= e^{-x} (2x - x^2) + \frac{6x}{x^2 - 1} - \frac{1 - \ln(x)}{x^2}$$

2.

$$f(x) = (x^2 + 1)^3 e^{-x^2} + x \ln(e^x + 1)$$

Sol.

$$f'(x) = 3(x^2 + 1)^2 2x e^{-x^2} + (x^2 + 1)^3 (-2x e^{-x^2}) + \ln(e^x + 1) + x \frac{1}{e^x + 1} e^x$$

$$= 6x e^{-x^2} (x^2 + 1)^2 - 2x e^{-x^2} (x^2 + 1)^3 + \ln(e^x + 1) + x e^x \frac{1}{e^x + 1}$$

$$= 2x e^{-x^2} (x^2 + 1)^2 (2 - x^2) + \ln(e^x + 1) + x e^x \frac{1}{e^x + 1}$$

3.

$$f(x) = \frac{x^2 e^{-x}}{\ln(x)} + \ln(x^3 + 1) - \frac{1}{x^2 + 1}$$

Sol.

$$f'(x) = \frac{(2x e^{-x} - x^2 e^{-x}) \ln(x) - x^2 e^{-x} \frac{1}{x}}{(\ln(x))^2} + \frac{1}{x^3 + 1} 3x^2 + (x^2 + 1)^{-2} 2x$$

4.

$$f(x) = x^3 e^{-\frac{x^2}{2}} + \frac{\ln(x^2 + 1)}{x} - x \ln(x)$$

Sol.

$$f'(x) = 3x^2 e^{-\frac{x^2}{2}} + x^3 (-x) e^{-\frac{x^2}{2}} \frac{\frac{2x}{x^2+1} x + \ln(x^2 + 1)}{x^2} - (\ln(x) + 1)$$

5.

$$f(x) = (x^2 + 1)^2 e^{-x} + \frac{x \ln(x)}{x^2 + 1} + \ln(e^{-x^2} + 1)$$

Sol. $f'(x) = 2(x^2 + 1)2xe^{-x} + (x^2 + 1)^2(-e^{-x}) + \frac{(\ln(x) + \frac{1}{x}x)(x^2 + 1) - x \ln(x)2x}{(x^2 + 1)^2}$
